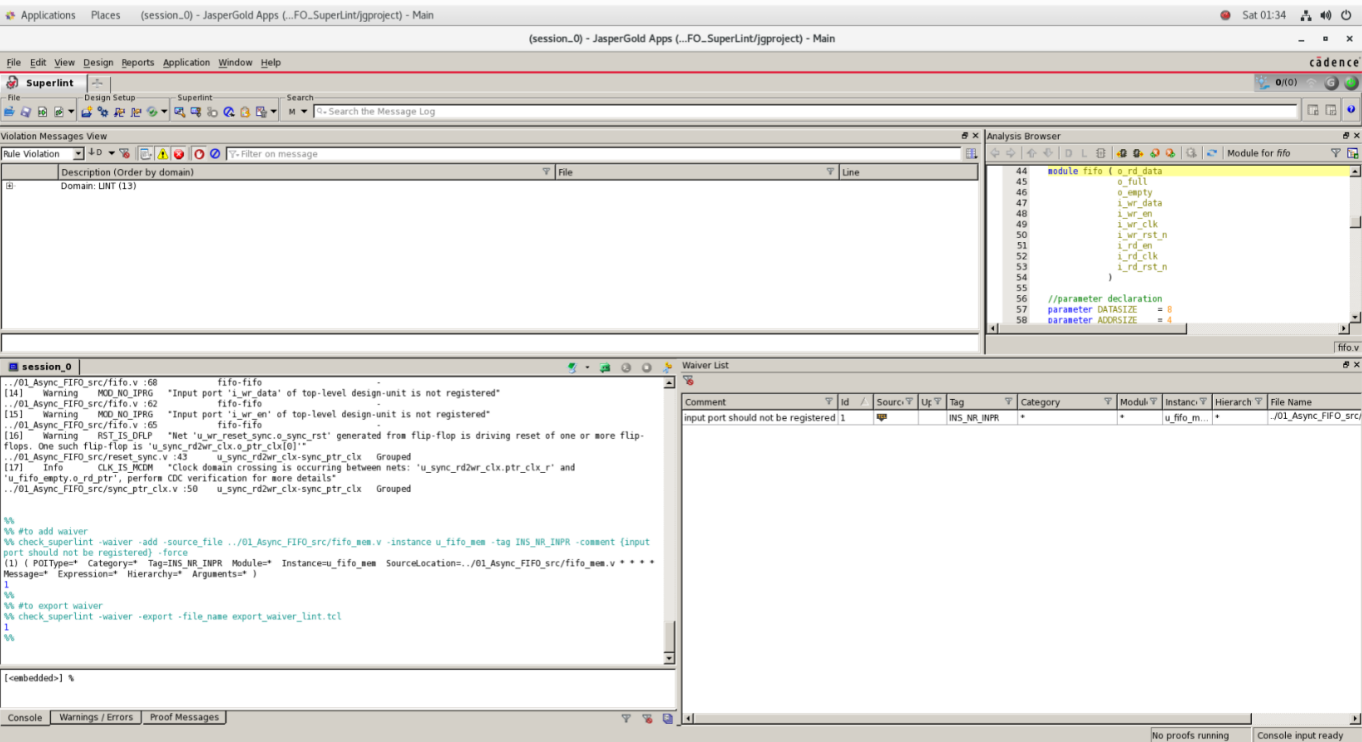
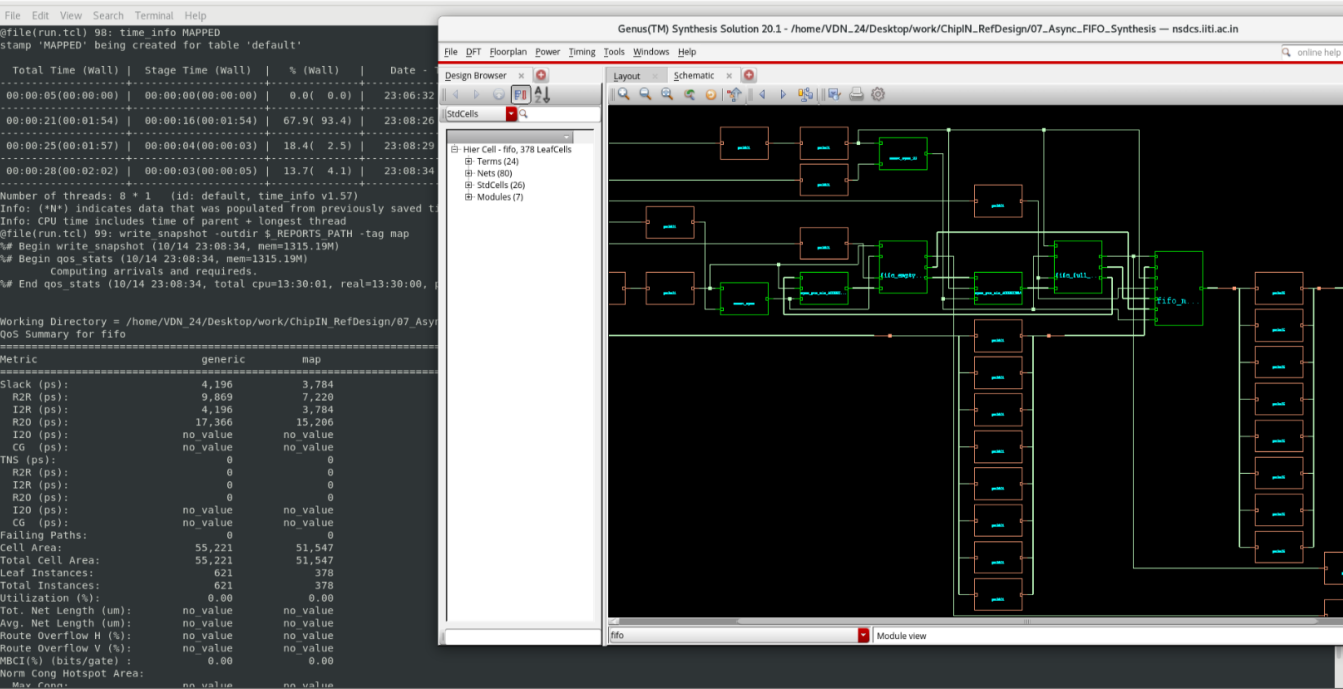


# FIFO Lint



# FIFO Synthesis



# FIFO Post Synthesis

File Edit View Search Terminal Help

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// Warning: This version is 1715 days old. You can download the latest version from [www.cadence.com](#)

// Check out Conformal Ultra 20.2 license

// Check out Conformal Asic 20.2 license

[VDN\_24@nsdcs 01\_RTL2Intermediate]\$ lec -xl rtl2intermed

// Warning: Using 'rtl2Intermediate.lec.do' as filename

CONFORMAL (R)

Version 20.20-s200 (01-Feb-2021) (64 bit)

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^C

[VDN\_24@nsdcs 01\_RTL2Intermediate]\$ cd ..

[VDN\_24@nsdcs 08\_Async\_FIFO\_Post\_Synthesis\_LEC]\$ cd 02\_1n

[VDN\_24@nsdcs 02\_intermediate2incremental]\$ lec -xl

fifo\_incremental.v

//

[VDN\_24@nsdcs 02\_intermediate2incremental]\$ lec -xl in

intermediate2final.lec.do intermediate2final.lec.log

[VDN\_24@nsdcs 02\_intermediate2incremental]\$ lec -xl inte

// Warning: Using 'intermediate2final.lec.do' as filename

CONFORMAL (R)

Version 20.20-s200 (01-Feb-2021) (64 bit)

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Conformal(R) Logic Equivalence Checking

File Compare Tools Custom Preferences Window Help

cadence

Design Library Tool Setup Design Data Rule Checking

Golden

Revised

fifo

26 library cells

u\_fifo\_empty(fifo\_empty\_ADDRSIZE4)

u\_fifo\_full(fifo\_full\_ADDRSIZE4)

u\_fifo\_mem(fifo\_mem\_DATASIZE8\_ADDRSIZE4\_MEM\_DEPTH16)

u\_rd\_reset\_sync(reset\_sync)

u\_sync\_rd2wr\_clk(sync\_ptr\_clk\_ADDRSIZE4)

u\_sync\_wir2rd\_clk(sync\_ptr\_clk\_ADDRSIZE4\_34)

u\_wr\_reset\_sync(reset\_sync\_33)

fifo

26 library cells

u\_fifo\_empty(fifo\_empty\_ADDRSIZE4)

u\_fifo\_full(fifo\_full\_ADDRSIZE4)

u\_fifo\_mem(fifo\_mem\_DATASIZE8\_ADDRSIZE4\_MEM\_DEPTH16)

u\_rd\_reset\_sync(reset\_sync)

u\_sync\_rd2wr\_clk(sync\_ptr\_clk\_ADDRSIZE4)

u\_sync\_wir2rd\_clk(sync\_ptr\_clk\_ADDRSIZE4\_34)

u\_wr\_reset\_sync(reset\_sync\_33)

Automatic setup finished.

// Command: report\_unmapped\_points -summary

// Command: report\_unmapped\_points -summary

There is no unmapped point

// Command: report\_unmapped\_points -notmapped

// Command: report\_unmapped\_points -notmapped

There is no unmapped point

// Command: add\_compared\_points -all

// Command: add\_compared\_points -all

182 compared points added to compare list

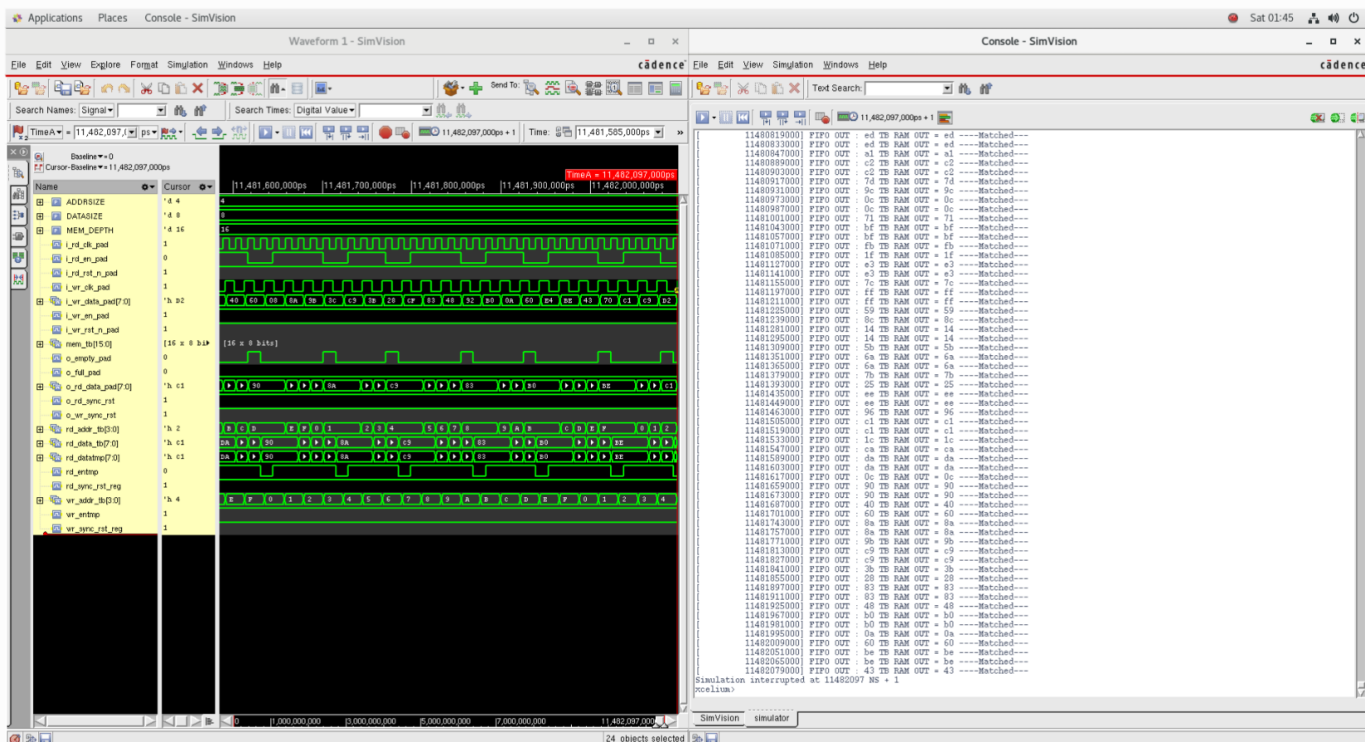
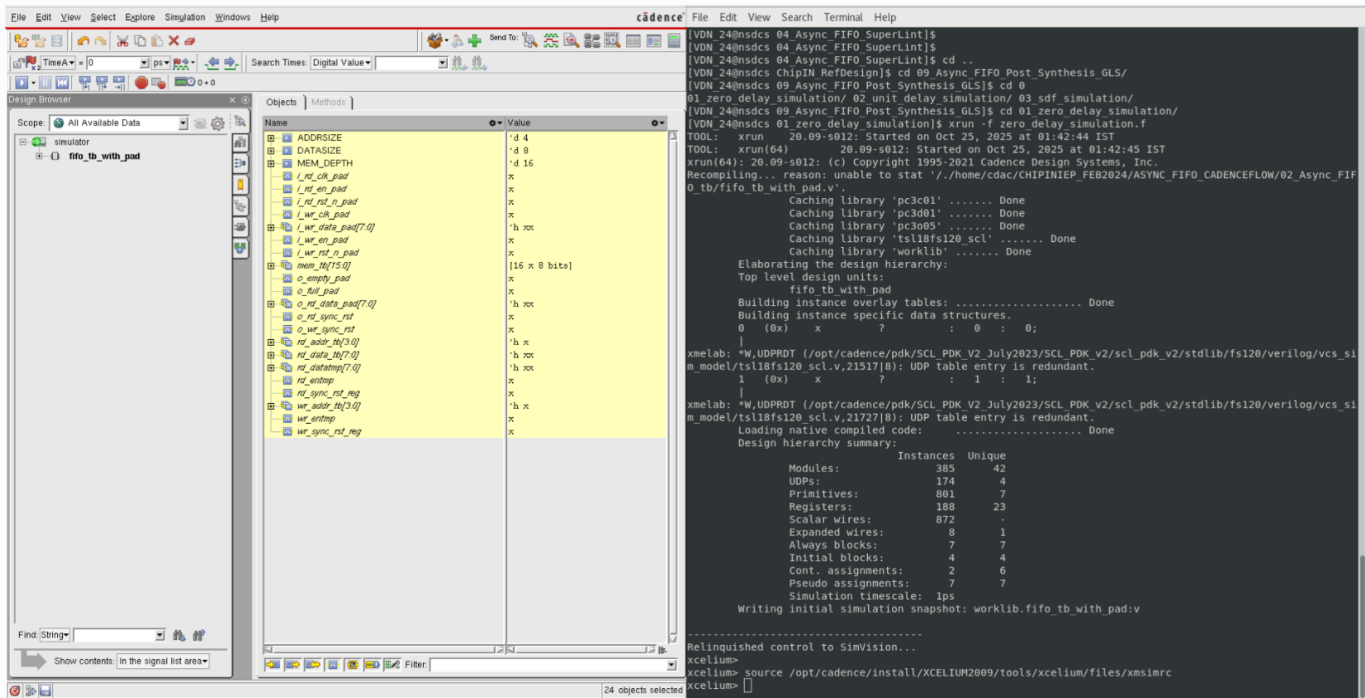
// Command: compare

// Command: compare

SETUP> dofile intermediate2final.lec.do

Comparing 168 out of 182 points, 0 Non-equivalent

## Zero\_delay\_simulation



# Unit\_delay Simulation

The top window shows the command-line interface of the simulator. It displays the process of building instance overlay tables, generating native compiled code, and building instance specific data structures. The output includes a list of modules and their instances, and a summary of the design hierarchy.

```
Top level design unit:
fifo_tb_with_pad
Building instance overlay tables: ..... Done
Generating native compiled code:
  tsl18fs120 scl.decrql:v <0x3a5653bf>
    streams: 0, words: 0
  tsl18fs120 scl.dfcrlb:v <0x7f2a8244>
    streams: 0, words: 0
  tsl18fs120 scl.oain22d1:v <0x6edd3fff>
    streams: 0, words: 0
  tsl18fs120 scl.dfrpb1:v <0x70d06af2>
    streams: 0, words: 0
  tsl18fs120 scl.ah01d1:v <0x3a2485f2>
    streams: 0, words: 0
  tsl18fs120 scl.xr02d1:v <0x3a866804>
    streams: 0, words: 0
  tsl18fs120 scl.aoin22d1:v <0x3796404a>
    streams: 0, words: 0
  tsl18fs120 scl.mx02d1:v <0x68c561f5>
    streams: 0, words: 0
  tsl18fs120 scl.dfcrlq:v <0x31f855bd>
    streams: 0, words: 0
Building instance specific data structures.
  0 (0x) x ? : 0 : 0;
xmelab: *W,UDPROT /opt/cadence/pdk/SCL_PDK_V2_July2023/SCL_PDK_V2/scl_pdk_v2/stdlib/fsl20/verilog/vcs_sl
m_model/tsl18fs120_scl.v.21517[B]: UDP Table entry is redundant.
  1 (0x) x ? : 1 : 1;
xmelab: *W,UDPROT /opt/cadence/pdk/SCL_PDK_V2_July2023/SCL_PDK_V2/scl_pdk_v2/stdlib/fsl20/verilog/vcs_sl
m_model/tsl18fs120_scl.v.21517[B]: UDP Table entry is redundant.
Loading native compiled code: ..... Done
Design hierarchy summary:
Modules: 395 42
UDPs: 174 4
Primitives: 801 7
Registers: 188 23
Scalar wires: 884 -
Expanded wires: 8 1
Always blocks: 7 7
Initial blocks: 4 4
Cont. assignments: 2 6
Pseudo assignments: 7 7
Simulation timescale: lps
Writing initial simulation snapshot: worklib.fifo_tb_with_pad.v
Relinquished control to SimVision...
xcclium> source /opt/cadence/install/XCCLIUM2009/tools/xcclium/files/xmsimrc
xcclium>
```

The bottom window shows the 'Objects' list in the simulator. It contains a table of objects and their values.

| Name               | Value         |
|--------------------|---------------|
| ADDSIZE            | 'd 4          |
| DATASIZE           | 'd 8          |
| MEM_DEPTH          | 'd 16         |
| lrd_clk_pad        | 0             |
| lrd_en_pad         | 0             |
| lrd_rst_n_pad      | 1             |
| lwr_clk_pad        | 1             |
| lwr_en_pad         | 'h 06         |
| lwr_rst_n_pad      | 1             |
| mem_b[15:0]        | [16 x 8 bits] |
| o_empty_pad        | 0             |
| o_full_pad         | 0             |
| o_rd_data_pad[7:0] | 'h E3         |
| o_rd_sync_rst      | 1             |
| o_wr_sync_rst      | 1             |
| rd_addr_b[3:0]     | 'h 4          |
| rd_data_b[7:0]     | 'h E3         |
| rd_data[7:0]       | 'h E3         |
| rd_rstnp           | 0             |
| rd_sync_rst_reg    | 1             |
| wr_addr_b[3:0]     | 'h 7          |
| wr_rstnp           | 1             |
| wr_sync_rst_reg    | 1             |

The top window shows the 'Signal' list in the simulator. It contains a table of signals and their values.

| Name               | Value         |
|--------------------|---------------|
| o_empty_pad        | 0             |
| ADDSIZE            | 'd 4          |
| DATASIZE           | 'd 8          |
| MEM_DEPTH          | 'd 16         |
| lrd_clk_pad        | 0             |
| lrd_en_pad         | 0             |
| lrd_rst_n_pad      | 1             |
| lwr_clk_pad        | 1             |
| lwr_en_pad         | 'h 06         |
| lwr_rst_n_pad      | 1             |
| mem_b[15:0]        | [16 x 8 bits] |
| o_full_pad         | 0             |
| o_rd_data_pad[7:0] | 'h E3         |
| o_rd_sync_rst      | 1             |
| o_wr_sync_rst      | 1             |
| rd_addr_b[3:0]     | 'h 4          |
| rd_data_b[7:0]     | 'h E3         |
| rd_data[7:0]       | 'h E3         |
| rd_rstnp           | 0             |
| rd_sync_rst_reg    | 1             |
| wr_addr_b[3:0]     | 'h 7          |
| wr_rstnp           | 1             |
| wr_sync_rst_reg    | 1             |

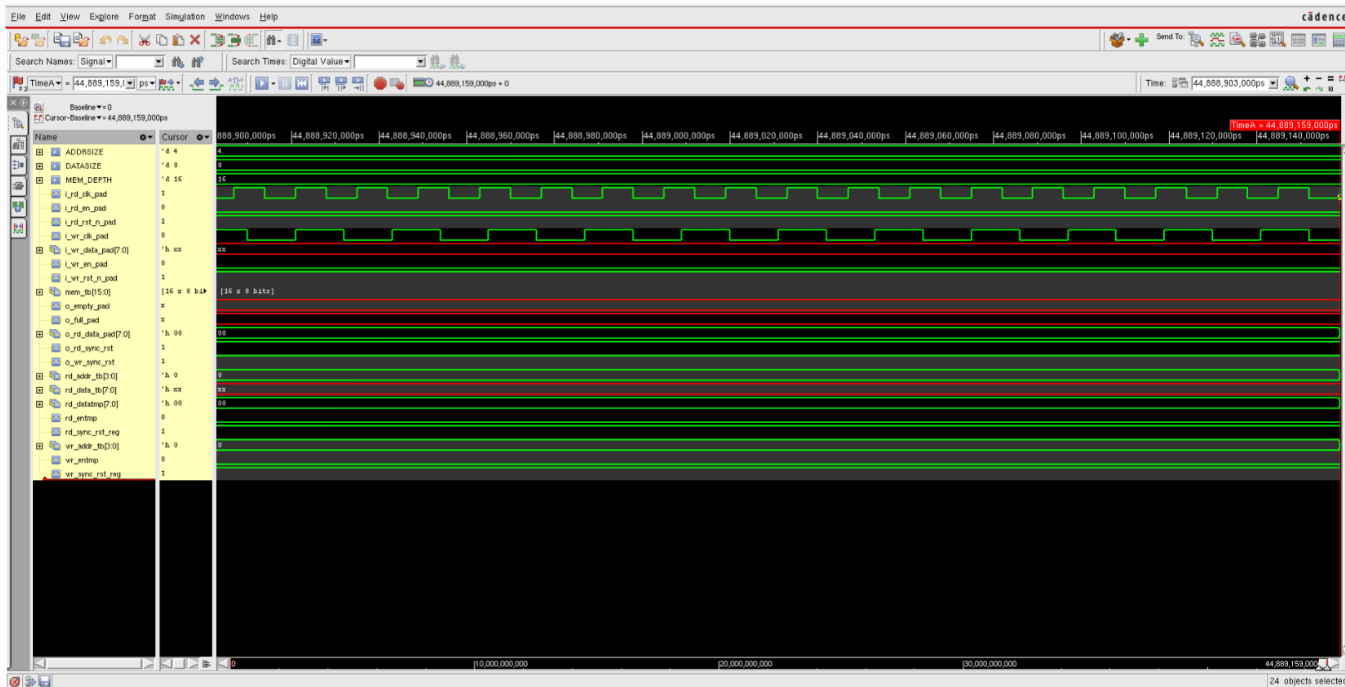
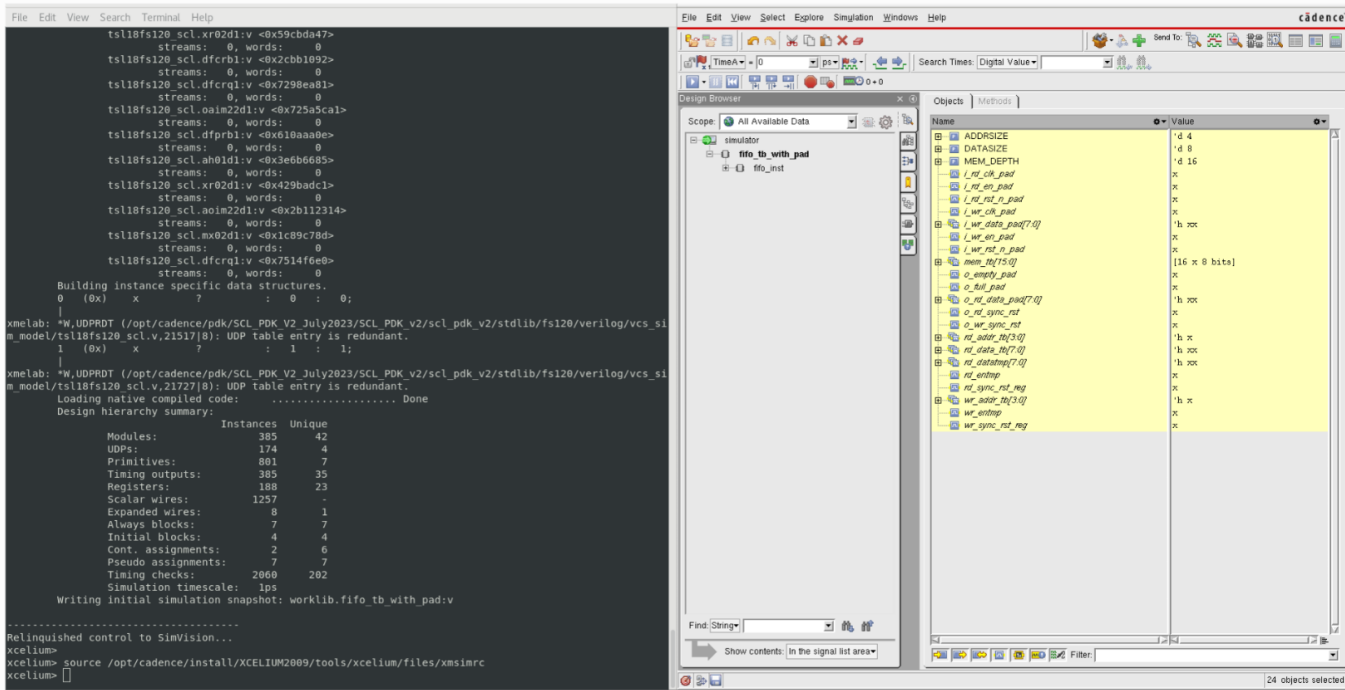
The bottom window shows a timing diagram with a cursor at 23,701,469.00ps. The diagram displays the signals listed in the 'Signal' list. The signals are color-coded: green for logic 0, red for logic 1, and blue for logic X. The diagram shows the signals' behavior over time, with a cursor indicating the current time point.

The right window shows a list of simulation errors. The errors are listed in a table with columns for the error type, the error message, and the error location.

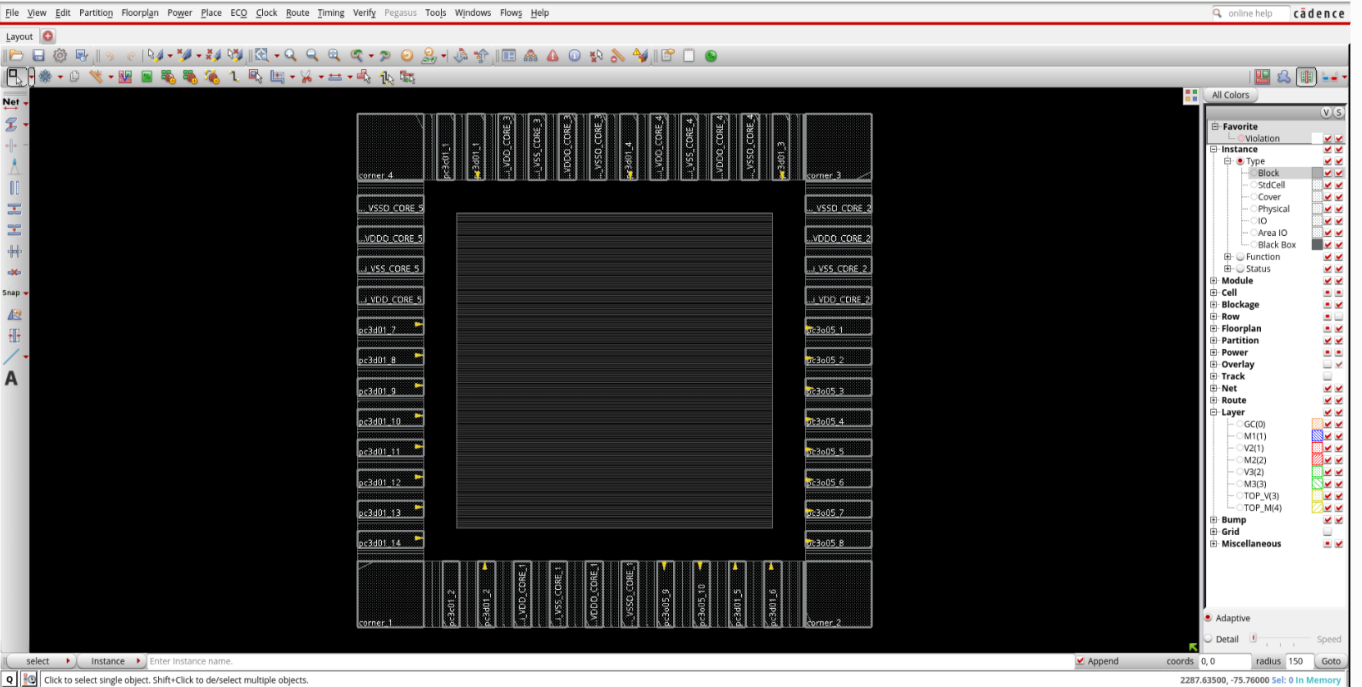
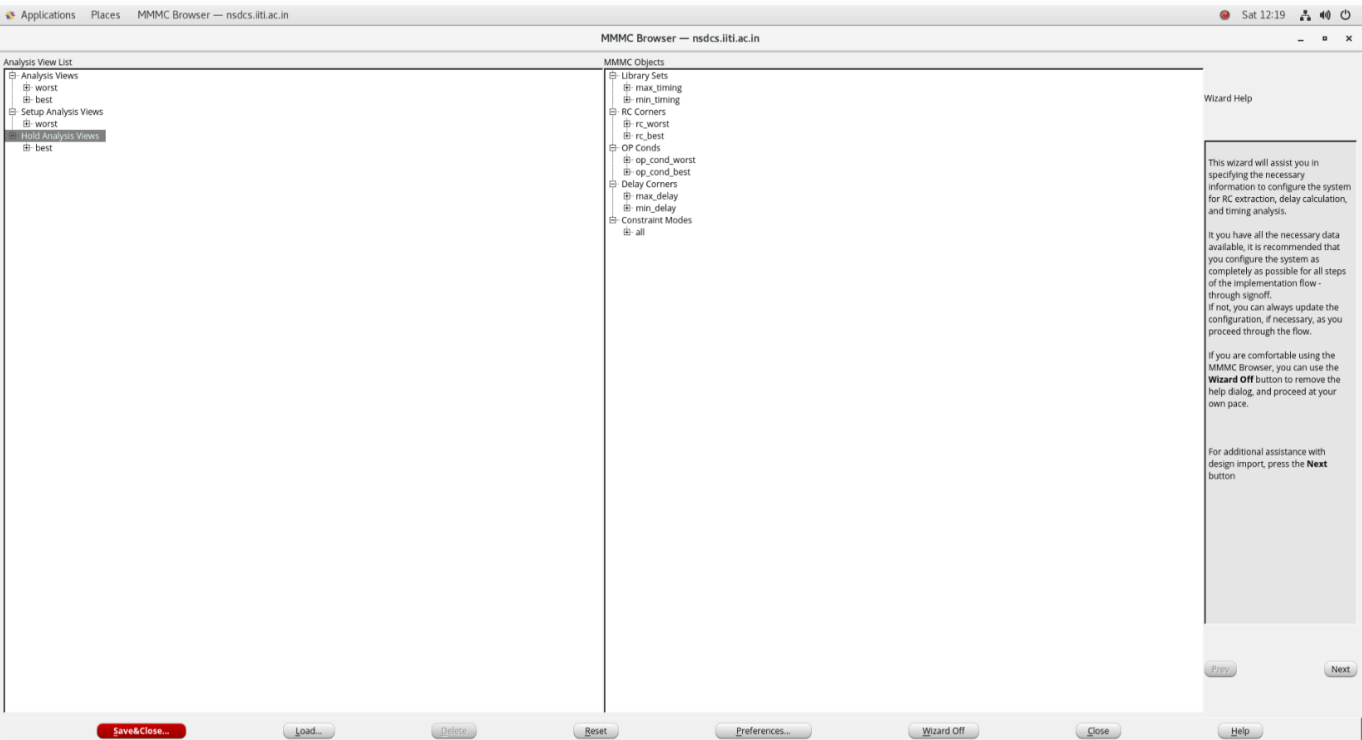
| Error Type | Error Message             | Error Location |
|------------|---------------------------|----------------|
| Matched    | 23700210000 FIFO OUT = b3 | Matched        |
| Matched    | 23700215000 FIFO OUT = 5a | Matched        |
| Matched    | 23700220000 FIFO OUT = 71 | Matched        |
| Matched    | 23700225000 FIFO OUT = 73 | Matched        |
| Matched    | 23700230000 FIFO OUT = c7 | Matched        |
| Matched    | 23700235000 FIFO OUT = c7 | Matched        |
| Matched    | 23700240000 FIFO OUT = 19 | Matched        |
| Matched    | 23700245000 FIFO OUT = 19 | Matched        |
| Matched    | 23700250000 FIFO OUT = 20 | Matched        |
| Matched    | 23700255000 FIFO OUT = 52 | Matched        |
| Matched    | 23700260000 FIFO OUT = 9a | Matched        |
| Matched    | 23700265000 FIFO OUT = 22 | Matched        |
| Matched    | 23700270000 FIFO OUT = 22 | Matched        |
| Matched    | 23700275000 FIFO OUT = 20 | Matched        |
| Matched    | 23700280000 FIFO OUT = a3 | Matched        |
| Matched    | 23700285000 FIFO OUT = 39 | Matched        |
| Matched    | 23700290000 FIFO OUT = a7 | Matched        |
| Matched    | 23700295000 FIFO OUT = 55 | Matched        |
| Matched    | 23700300000 FIFO OUT = 05 | Matched        |
| Matched    | 23700305000 FIFO OUT = 4f | Matched        |
| Matched    | 23700310000 FIFO OUT = f7 | Matched        |
| Matched    | 23700315000 FIFO OUT = b4 | Matched        |
| Matched    | 23700320000 FIFO OUT = 05 | Matched        |
| Matched    | 23700325000 FIFO OUT = c5 | Matched        |
| Matched    | 23700330000 FIFO OUT = 9b | Matched        |
| Matched    | 23700335000 FIFO OUT = 9b | Matched        |
| Matched    | 23700340000 FIFO OUT = da | Matched        |
| Matched    | 23700345000 FIFO OUT = 3a | Matched        |
| Matched    | 23700350000 FIFO OUT = d3 | Matched        |
| Matched    | 23700355000 FIFO OUT = d3 | Matched        |
| Matched    | 23700360000 FIFO OUT = e1 | Matched        |
| Matched    | 23700365000 FIFO OUT = e1 | Matched        |
| Matched    | 23700370000 FIFO OUT = cd | Matched        |
| Matched    | 23700375000 FIFO OUT = cd | Matched        |



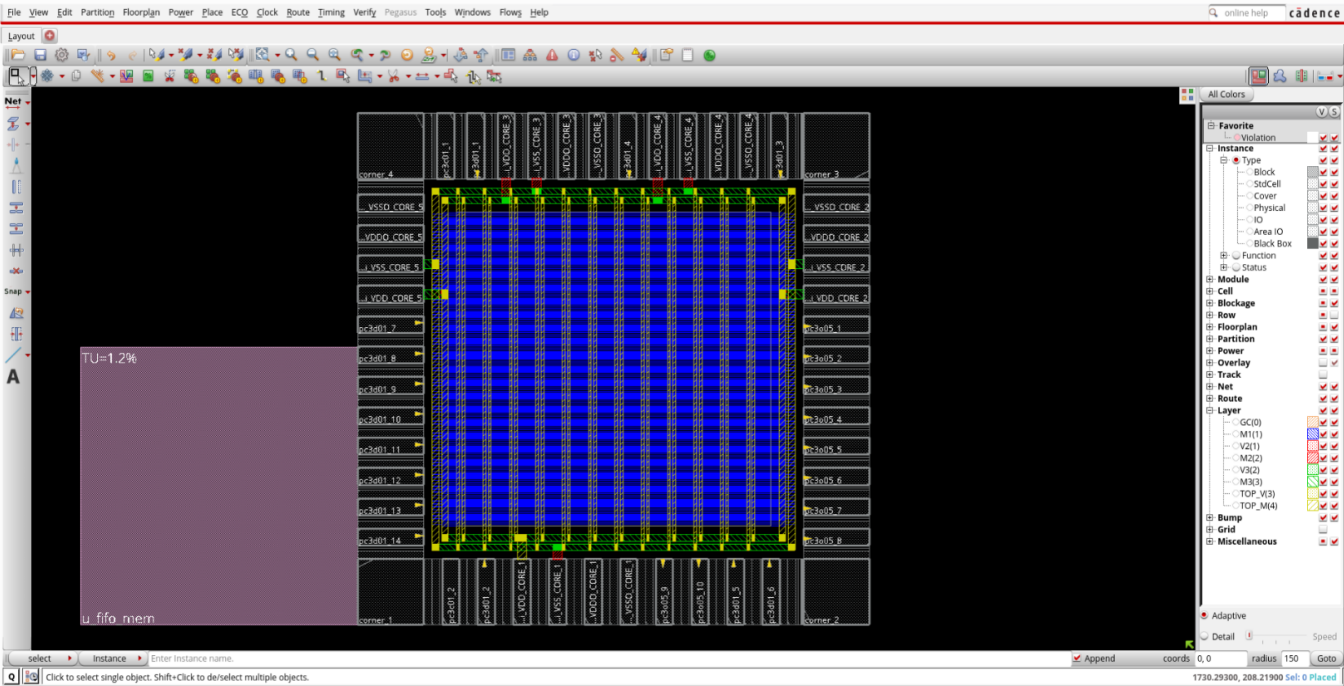
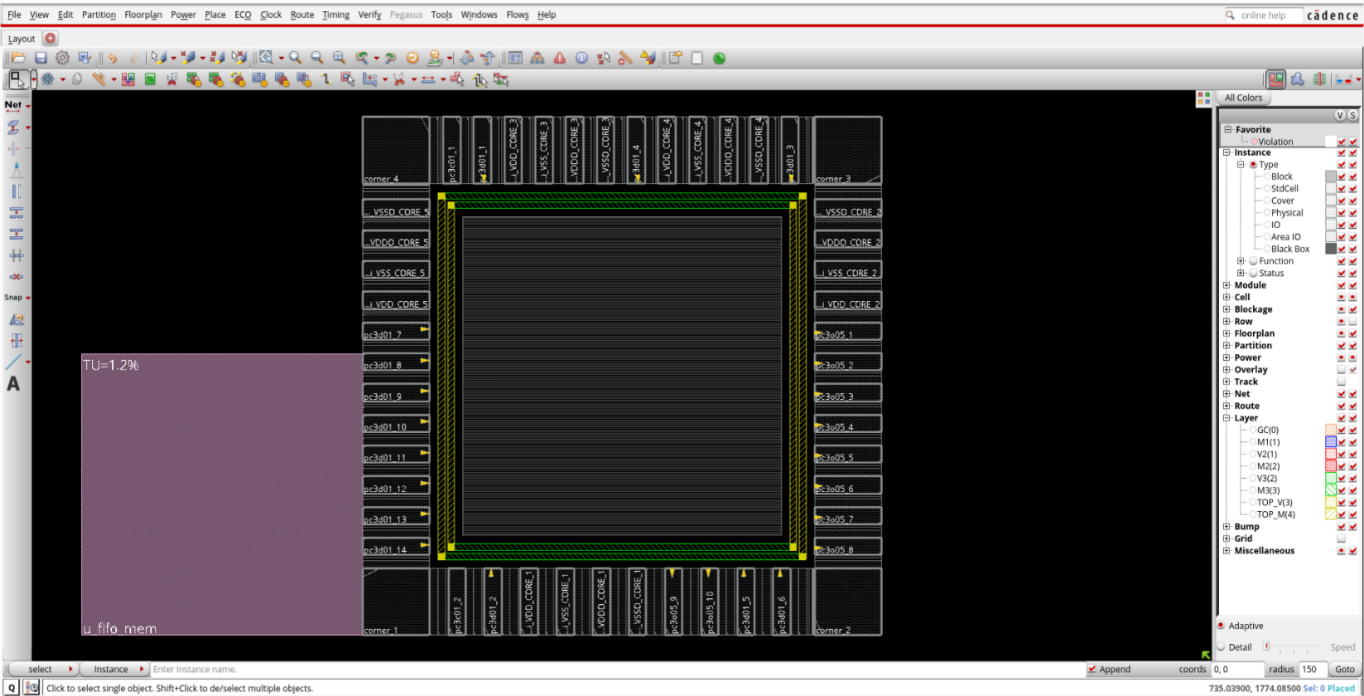
## Sdf\_simulation



# FIFO Floorplanning



# FIFO Power Planning



# FIFO Placement

```
File Edit View Search Terminal Help
**place_opt design ... cpu = 0:00:19, real = 0:00:26, mem = 1466.2M **
*** Finished GigaPlace ***

*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
WARNING IMPEXT-6166 3 Capacitance table file(s) without the EX...
WARNING IMPEXT-3530 3 The process node is not set. Use the com...
WARNING IMPSP-9025 1 No scan chain specified/traced.
*** Message Summary: 7 warning(s), 0 error(s)

*** place_opt design #1 [finish] : cpu/real = 0:00:19.2/0:00:25.9 (0.7), totSession cpu/real = 0:01:50.2/0:05:55.0 (0.3), mem = 1466.2M
innovus 1> *** timeDesign #2 [begin] : totSession cpu/real = 0:01:53.0/0:06:06.0 (0.3), mem = 1476.4M
Start to check current routing status for nets...
All nets are already routed correctly.
End to check current routing status for nets (mem=1466.4M)

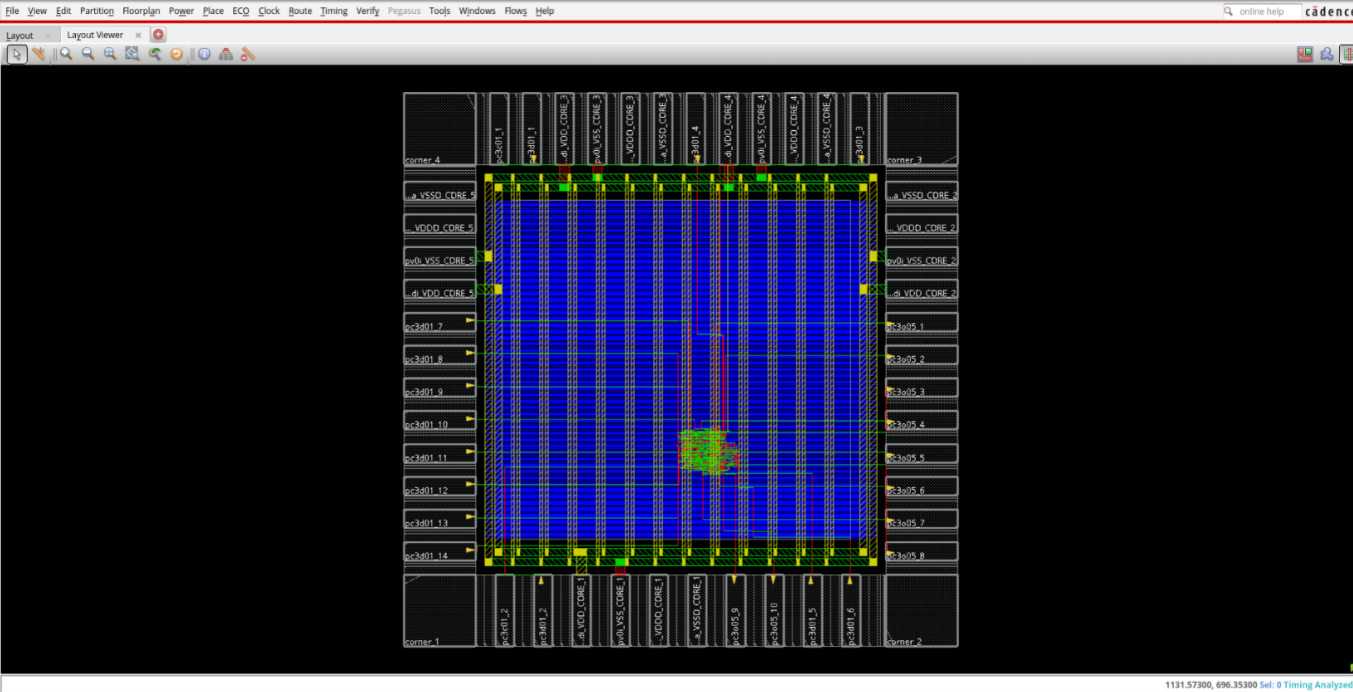
-----
timeDesign Summary
-----

Setup views included:
worst

-----
Setup mode | all | reg2reg | default |
-----
WNS (ns): | 5.203 | 10.016 | 5.203 |
TNS (ns): | 0.000 | 0.000 | 0.000 |
Violating Paths: | 0 | 0 | 0 |
All Paths: | 465 | 328 | 137 |
-----

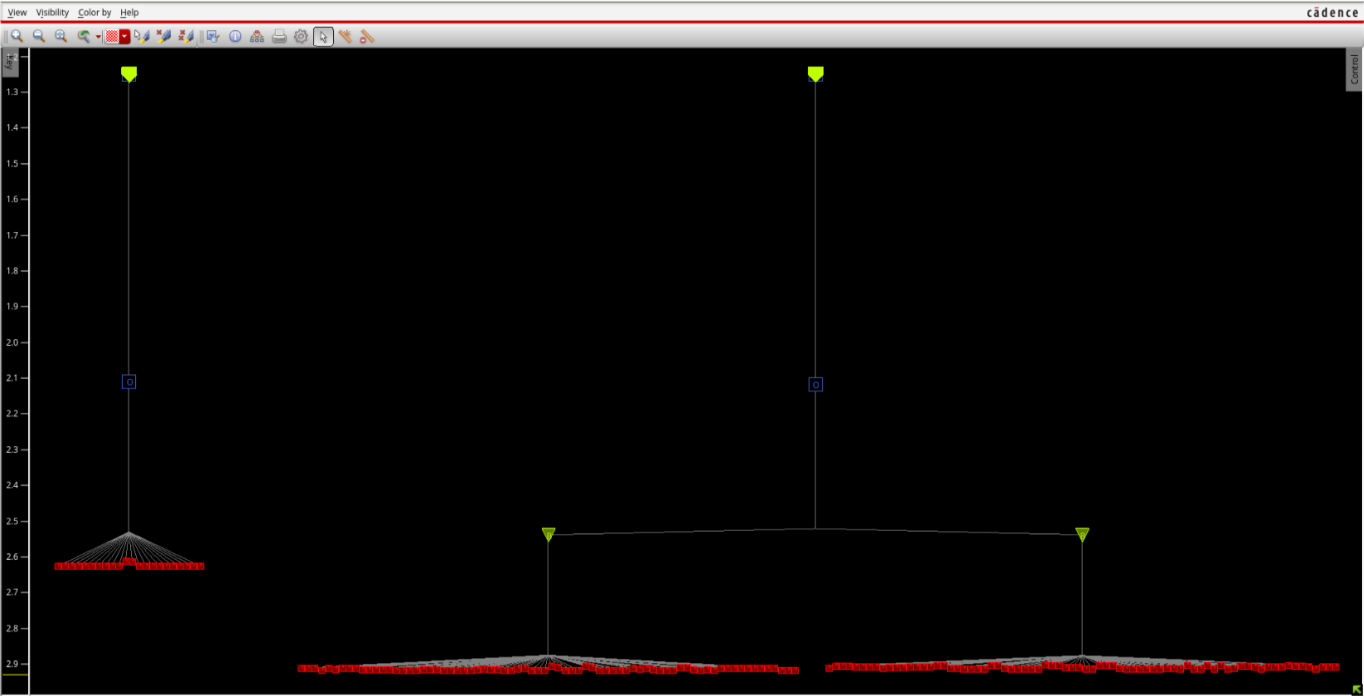
-----
DRVs | Real | Total |
-----
Nr nets(terms) | Worst Vio | Nr nets(terms) |
-----
max_cap | 0 (0) | 0.000 | 0 (0) |
max_tran | 0 (0) | 0.000 | 0 (0) |
max_fanout | 0 (0) | 0 | 0 (0) |
max_length | 0 (0) | 0 | 0 (0) |
-----

Density: 1.255%
Routing Overflow: 0.00% H and 0.00% V
-----
Reported timing to dir timingReports
Total CPU time: 0.52 sec
Total Real time: 12.0 sec
Total Memory Usage: 1475.167969 Mbytes
*** timeDesign #2 [finish] : cpu/real = 0:00:00.5/0:00:11.8 (0.0), totSession cpu/real = 0:01:53.5/0:06:18.6 (0.3), mem = 1475.2M
innovus 1> 
```



# FIFO Clock Tree synthesis

```
File Edit View Search Terminal Help
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=1628.77 CPU=0:00:00.2 REAL=0:00:01.0)
End delay calculation (fullDC). (MEM=1620.77 CPU=0:00:00.2 REAL=0:00:01.0)
#% Begin save design ... (date=10/28 19:49:45, mem=1267.4M)
% Begin Save ccopt configuration ... (date=10/28 19:49:45, mem=1267.4M)
% End Save ccopt configuration ... (date=10/28 19:49:45, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
% Begin Save netlist data ... (date=10/28 19:49:45, mem=1267.4M)
Writing Binary DB to ./ClockTreeSynthesis/ CTS.enc.dat/fifo.v.bin in single-threaded mode...
% End Save netlist data ... (date=10/28 19:49:45, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
Saving symbol-table file ...
Saving congestion map file ./ClockTreeSynthesis/ CTS.enc.dat/fifo.route.congmap.gz ...
% Begin Save AAE data ... (date=10/28 19:49:45, mem=1267.4M)
Saving AAE Data ...
% End Save AAE data ... (date=10/28 19:49:45, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
Saving preference file ./ClockTreeSynthesis/ CTS.enc.dat/gui.pref.tcl ...
Saving mode setting ...
Saving global file ...
% Begin Save floorplan data ... (date=10/28 19:49:45, mem=1267.4M)
Saving floorplan file ...
% End Save floorplan data ... (date=10/28 19:49:50, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
Saving PG file ./ClockTreeSynthesis/ CTS.enc.dat/fifo.pg.gz, version#2, (Created by Innovus v20.14-s095_1 on Tue Oct 28 19:49:51 2025)
*** Completed savePGFile (cpu=0:00:00.0 real=0:00:00.0 mem=1535.3M) ***
Saving Drc markers ...
... No Drc file written since there is no markers found.
% Begin Save placement data ... (date=10/28 19:49:51, mem=1267.4M)
** Saving stdCellPlacement binary (version# 2) ...
Save Adaptive View Pruning View Names to Binary file
% End Save placement data ... (date=10/28 19:49:51, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
% Begin Save routing data ... (date=10/28 19:49:51, mem=1267.4M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:00.0 mem=1535.3M) ***
% End Save routing data ... (date=10/28 19:49:51, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
Saving property file ./ClockTreeSynthesis/ CTS.enc.dat/fifo.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=1538.3M) ***
% Saving pin access data to file ./ClockTreeSynthesis/ CTS.enc.dat/fifo.apa ...
#
Saving rc congestion map ./ClockTreeSynthesis/ CTS.enc.dat/fifo.congmap.gz ...
% Begin Save power constraints data ... (date=10/28 19:49:51, mem=1267.4M)
% End Save power constraints data ... (date=10/28 19:49:51, total cpu=0:00:00.0, real=0:00:00.0, peak res=1267.4M, current mem=1267.4M)
Generated self-contained design CTS.enc.dat
#% End save design ... (date=10/28 19:49:52, total cpu=0:00:00.6, real=0:00:07.0, peak res=1267.4M, current mem=1267.4M)
*** Message Summary: 0 warning(s), 0 error(s)
0
Innovus 2> █
```



cost cts

```
File Edit View Search Terminal Help
# Design Stage: PreRoute
# Design Name: fifo
# Design Mode: 90nm
# Analysis Mode: MMMC OCV
# Parasitics Mode: No SPEF/RCDB
# Signoff Settings: SI Off
#####
Start delay calculation (fullDC) (1 T). (MEM=1598.12)
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=1618.34 CPU=0:00:00.1 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1618.34 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.1 real=0:00:00.0 totSessionCpu=0:02:07 mem=1618.3M)

-----
timeDesign Summary
-----

Setup views included:
Worst

+-----+-----+-----+-----+
| Setup mode | all | reg2reg | default |
+-----+-----+-----+-----+
| WNS (ns): | 4.908 | 8.779 | 4.908 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 465 | 328 | 137 |
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| DRV's | Real | Total |
| Nr nets(terms) | Worst Vio | Nr nets(terms) |
+-----+-----+-----+-----+
| max_cap | 0 (0) | 0.000 | 0 (0) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
| max_length | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+-----+

Density: 1.318%
Routing Overflow: 0.00% H and 0.00% V
-----
Reported timing to dir timingReports
Total CPU time: 0.73 sec
Total Real time: 7.0 sec
Total Memory Usage: 1591.597656 Mbytes
*** timeDesign #3 [finish] : cpu/real = 0:00:00.7/0:00:07.0 (0.1), totSession cpu/real = 0:02:07.1/0:07:10.0 (0.3), mem = 1591.6M
innovus 2> █
```

Hold voilation removed

```
File Edit View Search Terminal Help
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
End delay calculation. (MEM=1704.63 CPU=0:00:00.1 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1704.63 CPU=0:00:00.1 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.2 real=0:00:00.0 totSessionCpu=0:03:03 mem=1704.6M)

-----
optDesign Final Summary
-----

Setup views included:
Worst
Hold views included:
Best

+-----+-----+-----+-----+
| Setup mode | all | reg2reg | default |
+-----+-----+-----+-----+
| WNS (ns): | 4.908 | 8.779 | 4.908 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 465 | 328 | 137 |
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| Hold mode | all | reg2reg | default |
+-----+-----+-----+-----+
| WNS (ns): | 0.251 | 0.251 | 0.638 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 465 | 328 | 137 |
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| DRV's | Real | Total |
| Nr nets(terms) | Worst Vio | Nr nets(terms) |
+-----+-----+-----+-----+
| max_cap | 0 (0) | 0.000 | 0 (0) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
| max_length | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+-----+

Density: 1.319%
Routing Overflow: 0.00% H and 0.00% V
-----
**optDesign ... cpu = 0:00:05, real = 0:00:12, mem = 1414.0M, totSessionCpu=0:03:03 **
*** Finished optDesign ***
Info: Destroy the CCOpt slew target map.
*** optDesign #3 [finish] : cpu/real = 0:00:05.1/0:00:11.6 (0.4), totSession cpu/real = 0:03:03.5/0:10:41.5 (0.3), mem = 1672.9M
innovus 2> █
```



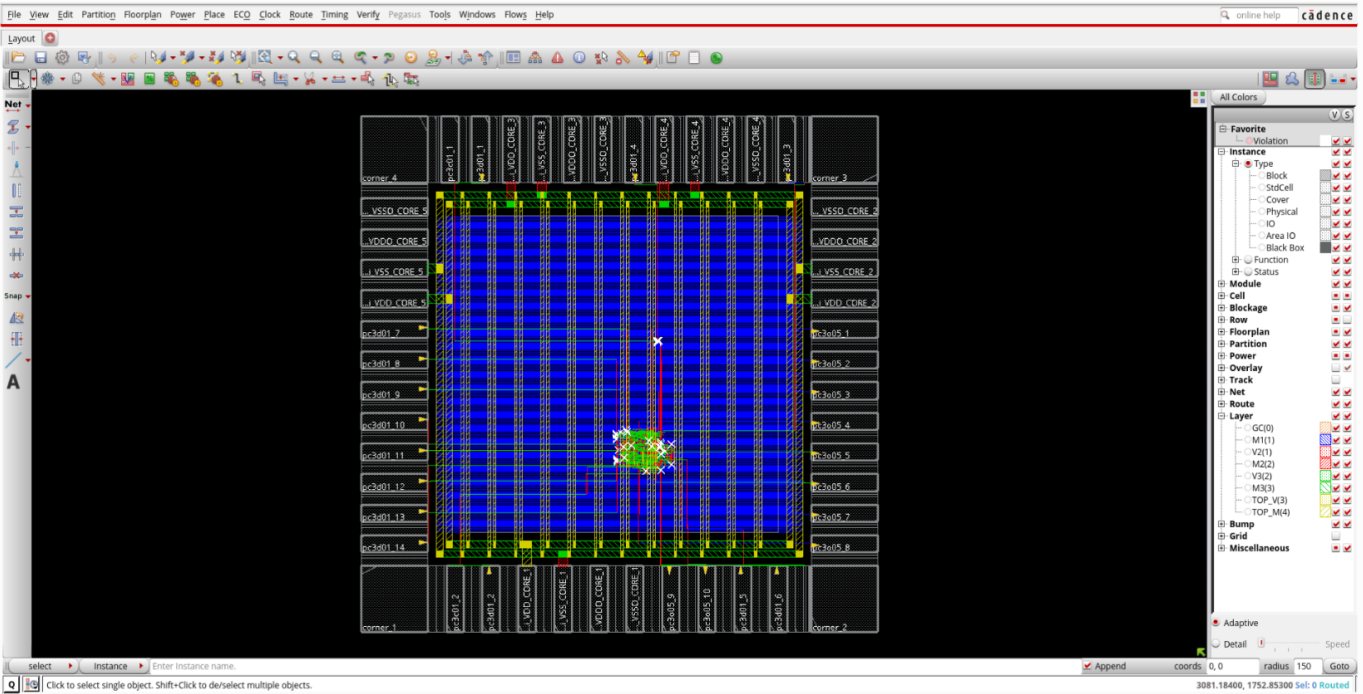
# FIFO\_ Routing

```
File Edit View Search Terminal Help

#Total number of net violated process antenna rule = 8
#Total number of violations on LAYER M1 = 22
#Total number of violations on LAYER M2 = 6
#Total number of violations on LAYER M3 = 0
#Total number of violations on LAYER TOP_M = 0
#Post Route wire spread is done.
#Total wire length = 44458 um.
#Total half perimeter of net bounding box = 37669 um.
#Total wire length on LAYER M1 = 6273 um.
#Total wire length on LAYER M2 = 18851 um.
#Total wire length on LAYER M3 = 18947 um.
#Total wire length on LAYER TOP_M = 387 um.
#Total number of vias = 2725
#Up-Via Summary (total 2725):
#
#-----
# M1          1641
# M2          1062
# M3           22
#-----
#          2725
#
#DetailRoute Statistics:
#Cpu time = 00:00:05
#Elapsed time = 00:00:05
#Increased memory = 7.86 (MB)
#Total memory = 1098.59 (MB)
#Peak memory = 1161.74 (MB)
#
#GlobalDetailRoute statistics:
#Cpu time = 00:00:07
#Elapsed time = 00:00:07
#Increased memory = 6.11 (MB)
#Total memory = 1106.01 (MB)
#Peak memory = 1161.74 (MB)
#Number of warnings = 69
#Total number of warnings = 73
#Number of fails = 0
#Total number of fails = 0
#Complete GlobalDetailRoute on Thu Oct 30 18:20:40 2025
#
#Default setup view is reset to worst.
#Default setup view is reset to worst.
#RouteDesign: cpu time = 00:00:07, elapsed time = 00:00:07, memory = 1100.77 (MB), peak = 1161.74 (MB)

*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
WARNING IMPEXT-6166 1 Capacitance table file(s) without the EX...
WARNING IMPEXT-3530 1 The process node is not set. Use the com...
*** Message Summary: 2 warning(s), 0 error(s)

innova 1>
```



# Post Route Timing

```
File Edit View Search Terminal Help
Loading CTE timing window is completed (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 1418.6M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=1384.75)
Glitch Analysis: View worst -- Total Number of Nets Skipped = 0.
Glitch Analysis: View worst -- Total Number of Nets Analyzed = 0.
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 0.7 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 0.7 percent of the nets selected for SI analysis
End delay calculation. (MEM=1433.97 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1433.97 CPU=0:00:00.0 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.6 real=0:00:01.0 totSessionCpu=0:04:18 mem=1434.0M)

-----
timeDesign Summary
-----

Setup views included:
worst

-----+-----+-----+-----+
| Setup mode | all | reg2reg | default |
+-----+-----+-----+-----+
| WNS (ns): | 4.890 | 8.715 | 4.890 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 465 | 328 | 137 |
+-----+-----+-----+-----+

-----+-----+-----+-----+
| DRVs | Real | Total |
|-----+-----+-----+-----+
| | Nr nets(terms) | Worst Vio | Nr nets(terms) |
+-----+-----+-----+-----+
| max_cap | 0 (0) | 0.000 | 0 (0) |
| max_tran | 0 (0) | 0.000 | 0 (0) |
| max_fanout | 0 (0) | 0 | 0 (0) |
| max_length | 0 (0) | 0 | 0 (0) |
+-----+-----+-----+-----+

Density: 1.319%
Total number of glitch violations: 0
-----
Reported timing to dir timingReports
Total CPU time: 1.0 sec
Total Real time: 13.0 sec
Total Memory Usage: 1423.984375 Mbytes
Reset AAE Options
*** timeDesign #1 [finish] : cpu/real = 0:00:01.6/0:00:13.5 (0.1), totSession cpu/real = 0:04:18.2/0:16:31.4 (0.3), mem = 1424.0M
innovus 1> █
```

# Post route hold timing

```
File Edit View Search Terminal Help
*** Calculating scaling factor for min timing libraries using the default operating condition of each library.
Initializing multi-corner resistance Tables ...
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 98.2 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 98.2 percent of the nets selected for SI analysis
End delay calculation. (MEM=1432.34 CPU=0:00:00.1 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1432.34 CPU=0:00:00.1 REAL=0:00:00.0)
Loading CTE timing window with TwFlowType 0..(CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 1432.3M)
Add other clocks and setupctetOAAEClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 1432.3M)
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=1393.46)
Glitch Analysis: View best -- Total Number of Nets Skipped = 0.
Glitch Analysis: View best -- Total Number of Nets Analyzed = 0.
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 0.0 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 0.0 percent of the nets selected for SI analysis
End delay calculation. (MEM=1454.71 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1454.71 CPU=0:00:00.0 REAL=0:00:00.0)
*** Done Building Timing Graph (cpu=0:00:00.3 real=0:00:00.0 totSessionCpu=0:04:34 mem=1454.7M)

-----
timeDesign Summary
-----

Hold views included:
best

-----+-----+-----+-----+
| Hold mode | all | reg2reg | default |
+-----+-----+-----+-----+
| WNS (ns): | 0.253 | 0.253 | 0.645 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 465 | 328 | 137 |
+-----+-----+-----+-----+

Density: 1.319%
-----
Reported timing to dir timingReports
Total CPU time: 1.32 sec
Total Real time: 2.0 sec
Total Memory Usage: 1362.976562 Mbytes
Reset AAE Options
*** timeDesign #2 [finish] : cpu/real = 0:00:01.3/0:00:01.9 (0.7), totSession cpu/real = 0:04:34.2/0:17:34.7 (0.3), mem = 1363.0M
innovus 1> █
```

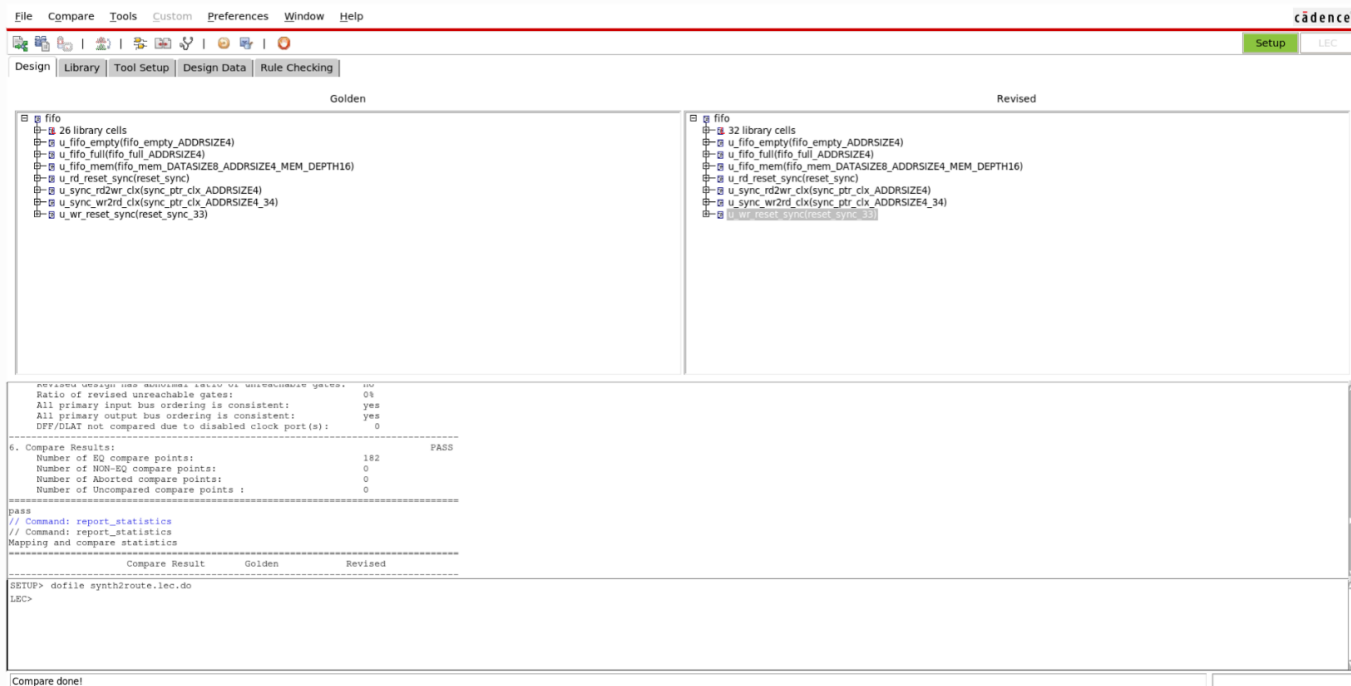
# Fillers

```
File Edit View Search Terminal Help
# Analysis Mode: MMWC OCV
# Parasitics Mode: SPEF/RCDB
# Signoff Settings: SI On
#####
AAE INFO: 1 threads acquired from CTE.
Start delay calculation (fullDC) (I T). (MEM=1512.03)
*** calculating scaling factor for max timing libraries using the default operating condition of each library.
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 98.2 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 98.2 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 98.2 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 98.2 percent of the nets selected for SI analysis
End delay calculation. (MEM=1540.78 CPU=0:00:00.3 REAL=0:00:00.0)
Loading CTE timing window with TopLevelType 0.-(CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 1540.8M)
Add other clocks and setupCtetoAAEClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.0, REAL = 0:00:00.0, MEM = 1540.8M)
Starting SI iteration 2
Start delay calculation (fullDC) (I T). (MEM=1508.99)
Glitch Analysis: View worst -- Total Number of Nets Skipped = 0.
Glitch Analysis: View worst -- Total Number of Nets Analyzed = 0.
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 0.7 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 0.7 percent of the nets selected for SI analysis
Glitch Analysis: View best -- Total Number of Nets Skipped = 0.
Glitch Analysis: View best -- Total Number of Nets Analyzed = 0.
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 0.7 percent of the nets selected for SI analysis
Total number of fetched objects 435
AAE INFO: Total number of nets for which stage creation was skipped for all views 0
AAE INFO-618: Total number of nets in the design is 441, 0.7 percent of the nets selected for SI analysis
End delay calculation. (MEM=1559.2 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=1559.2 CPU=0:00:00.0 REAL=0:00:01.0)
innovus 8> rcOut -spef ./Routing/fifo_postRoute.spef
RC Out has the following PVT Info:
RC-typical
Dumping Spef file.....
Printing 0 NET...
rcOut completed.: 100 %
RC Out from RCDB Completed (CPU Time= 0:00:00.1 MEM= 1573.2M)
innovus 9> □
```

```
File Edit View Search Terminal Help
so it should only be saved when it is really desired.
#% Begin save design ... (date=10/30 18:28:47, mem=1132.9M)
% Begin Save ccopt configuration ... (date=10/30 18:28:47, mem=1132.9M)
% End Save ccopt configuration ... (date=10/30 18:28:47, total cpu=0:00:00.0, real=0:00:00.0, peak res=1132.9M, current mem=1132.9M)
% Begin Save netlist data ... (date=10/30 18:28:47, mem=1132.9M)
Writing Binary DB to Routing/fifo_detailedRoute.enc.dat.tmp/fifo.v.bin in single-threaded mode...
% End Save netlist data ... (date=10/30 18:28:47, total cpu=0:00:00.0, real=0:00:00.0, peak res=1132.9M, current mem=1132.9M)
Saving symbol-table file ...
Saving congestion map file Routing/fifo_detailedRoute.enc.dat.tmp/fifo.route.congmap.gz ...
% Begin Save AAE data ... (date=10/30 18:28:47, mem=1132.9M)
Saving AAE Data ...
% End Save AAE data ... (date=10/30 18:28:47, total cpu=0:00:00.0, real=0:00:00.0, peak res=1132.9M, current mem=1132.9M)
Saving preference file Routing/fifo_detailedRoute.enc.dat.tmp/gui.pref.tcl ...
Saving mode setting ...
Saving global file ...
% Begin Save floorplan data ... (date=10/30 18:28:52, mem=1132.9M)
Saving floorplan file ...
% End Save floorplan data ... (date=10/30 18:28:53, total cpu=0:00:00.0, real=0:00:01.0, peak res=1132.9M, current mem=1132.9M)
Saving Pg file Routing/fifo_detailedRoute.enc.dat.tmp/fifo.pg.gz, version#2, (created by Innovus v20.14-s095_1 on Thu Oct 30 18:28:53 2025)
*** Completed savePgFile (cpu=0:00:00.0 real=0:00:00.0 mem=1422.4M) ***
Saving Drc markers ...
... 48 markers are saved ...
... 28 geometry drc markers are saved ...
... 20 antenna drc markers are saved ...
% Begin Save placement data ... (date=10/30 18:28:53, mem=1132.9M)
** Saving stdCellPlacement binary (version# 2) ...
Save Adaptive View Pruning View Names to Binary file
% End Save placement data ... (date=10/30 18:28:53, total cpu=0:00:00.0, real=0:00:00.0, peak res=1132.9M, current mem=1132.9M)
% Begin Save routing data ... (date=10/30 18:28:53, mem=1132.9M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:05.0 mem=1422.4M) ***
% End Save routing data ... (date=10/30 18:28:58, total cpu=0:00:00.0, real=0:00:05.0, peak res=1132.9M, current mem=1132.9M)
Saving property file Routing/fifo_detailedRoute.enc.dat.tmp/fifo.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=1425.4M) ***
#Saving pin access data to file Routing/fifo_detailedRoute.enc.dat.tmp/fifo.apa ...
#
% Begin Save power constraints data ... (date=10/30 18:28:58, mem=1132.9M)
% End Save power constraints data ... (date=10/30 18:28:58, total cpu=0:00:00.0, real=0:00:00.0, peak res=1132.9M, current mem=1132.9M)
Generated self-contained design fifo_detailedRoute.enc.dat.tmp
#% End save design ... (date=10/30 18:28:59, total cpu=0:00:00.6, real=0:00:12.0, peak res=1132.9M, current mem=1132.9M)
*** Message Summary: 0 warning(s), 0 error(s)

**WARN: (IMPSP-5217): addFiller command is running on a postRoute database. It is recommended to be followed by ecoRoute -target command to make the DRC clean.
Type 'man IMPSP-5217' for more detail.
*INFO: Adding fillers to top-module.
*INFO: Added 49198 filler insts (cell feedth3 / prefix FILLER).
*INFO: Added 427 filler insts (cell feedth3 / prefix FILLER).
*INFO: Added 568 filler insts (cell feedth / prefix FILLER).
*INFO: Swapped 0 special filler inst.
*INFO: Total 50185 filler insts added - prefix FILLER (CPU: 0:00:00.0).
For 50185 new insts, *** Applied 3 GNC rules (cpu = 0:00:00.0)
innovus 1> □
```

# FIFO Post Route LEC



## RC Signoff

```

File Edit View Search Terminal Help
**WARN: (IMPEXT-3505): setDelayCapMode -eng copyNetPropNet true (default=false) will be obsoleted along with its szgs2et equivalent. This parameter will continue to be supported in the current release, but it will be removed in the next major release of the software.
**WARN: (IMPCTE-107): The following globals have been obsoleted since version 20.14-s095.1. They will be removed in the next release.
timing.enable default delay_arc
Innovus I> Performing RC Extraction ...
Extraction called for design 'fifo' of instances=50788 and nets=441 using extraction engine 'postRoute' at effort level 'low'.
**WARN: (IMPEXT-3530): The process node is not set. Use the command setDesignMode -process <process node> prior to extraction for maximum accuracy and optimal automatic threshold setting.
Type 'man IMPEXT-3530' for more detail.
PostRoute (effortLevel low) RC Extraction called for design fifo.
RC Extraction called in multi-corner(2) mode.
**WARN: (IMPEXT-6166): Capacitance table file(s) without the EXTENDED section is being used for RC extraction. This is not recommended because it results in lower accuracy for rcg nets in preRoute extraction and for all nets in postRoute extraction using -effortLevel low. Regenerate capacitance table file(s) using the generateCapTbl command.
Type 'man IMPEXT-6166' for more detail.
Process corner(s) are loaded.
Corner: rc worst
Corner: rc best
extractDetailRC Option : -outfile /tmp/innovus temp 12556 nsdcs.i1t1.ac.in VDN 24.25xa3j/fifo 12556 LcpMbv.rcdb.d -basic
RC Mode: PostRoute -effortLevel low [Basic CapTable, RC Table Resistances]
RC Corner Indexes : 0 1
Capacitance Scaling Factor : 1.00000 1.00000
Coupling Cap. Scaling Factor : 1.00000 1.00000
Resistance Scaling Factor : 1.00000 1.00000
Clock Cap. Scaling Factor : 1.00000 1.00000
Clock Res. Scaling Factor : 1.00000 1.00000
Shrink Factor : 1.00000
Initializing multi-corner resistance tables ...
Checking LVS Completed (CPU Time= 0:00:00.0 MEM= 1321.8M)
Extracted 10.0298% (CPU Time= 0:00:00.0 MEM= 1385.9M)
Extracted 20.0397% (CPU Time= 0:00:00.0 MEM= 1385.9M)
Extracted 30.0298% (CPU Time= 0:00:00.0 MEM= 1385.9M)
Extracted 40.0397% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 50.0298% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 60.0397% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 70.0298% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 80.0397% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 90.0298% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 100% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Number of Extracted Resistors : 8105
Number of Extracted Ground Cap. : 8322
Number of Extracted Coupling Cap. : 14648
Filtering Xcap in 'relativeOnly' mode using values relative_c_threshold=0.03 and total_c_threshold=5ff.
Corner: rc worst
Corner: rc best
Checking LVS Completed (CPU Time= 0:00:00.0 MEM= 1369.9M)
PostRoute (effortLevel low) RC Extraction DONE (CPU Time: 0:00:00.4 Real Time: 0:00:01.0 MEM: 1369.859M)
RC Out has the following PVT info:
RC:rc best, Operating temperature -40 C
Dumping Spf file.....
Printing D.NET...
rcOut completed: 100 %
RC Out from RCDB Completed (CPU Times= 0:00:00.0 MEM= 1367.9M)
Innovus I>

```

# verify Connectivity

```
File Edit View Search Terminal Help
Extracted 30.0298% (CPU Time= 0:00:00.0 MEM= 1385.9M)
Extracted 40.0397% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 50.0298% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 60.0397% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 70.0298% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 80.0397% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 90.0298% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Extracted 100% (CPU Time= 0:00:00.1 MEM= 1385.9M)
Number of Extracted Resistors : 8105
Number of Extracted Ground Cap. : 8322
Number of Extracted Coupling Cap. : 14648
Filtering XCap in 'relativeOnly' mode using values relative_c_threshold=0.03 and total_c_threshold=5ff.
Corner: rc_worst
Corner: rc_best
Checking LVS Completed (CPU Time= 0:00:00.0 MEM= 1369.9M)
PostRoute (effortLevel low) RC Extraction DONE (CPU Time: 0:00:00.4 Real Time: 0:00:01.0 MEM: 1369.859M)
RC Out has the following PVT Info:
RC:rc_best, Operating temperature -40 c
Dumping SpEf file.....
Printing D_NET...
rcOut completed: 100 %
RC Out from RCDB Completed (CPU Time= 0:00:00.0 MEM= 1367.9M)
innovus l> VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY_CONNECTIVITY *****
Start Time: Thu Oct 30 19:25:48 2025

Design Name: fifo
Database Units: 1000
Design Boundary: (0.0000, 0.0000) (1939.8400, 1939.8400)
Error Limit = 1000; Warning Limit = 50
Check all nets
Net VDD_CORE: has an unconnected terminal, has special routes with opens.
Net VSS_CORE: has an unconnected terminal, has special routes with opens.
Net VDD0_CORE: no routing.
Net VSS0_CORE: no routing.

Begin Summary
  2 Problem(s) (IMPVFC-90): Net has no global routing and no special routing.
  82 Problem(s) (IMPVFC-90): Terminal(s) are not connected.
  2 Problem(s) (IMPVFC-200): Special Wires: Pieces of the net are not connected together.
  86 total info(s) created.
End Summary

End Time: Thu Oct 30 19:25:49 2025
Time Elapsed: 0:00:01.0

***** End: VERIFY_CONNECTIVITY *****
Verification Complete : 86 Viols.  0 Wrngs.
(CPU Time: 0:00:00.2 MEM: 0.000M)

innovus l> □
```

# DRC

```
File Edit View Search Terminal Help
VERIFY DRC ..... Sub-Area : 45 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1214.400 1214.400 1457.280 1457.280} 46 of 64
VERIFY DRC ..... Sub-Area : 46 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1457.280 1214.400 1700.160 1457.280} 47 of 64
VERIFY DRC ..... Sub-Area : 47 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1700.160 1214.400 1939.840 1457.280} 48 of 64
VERIFY DRC ..... Sub-Area : 48 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 1457.280 242.880 1700.160} 49 of 64
VERIFY DRC ..... Sub-Area : 49 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {242.880 1457.280 485.760 1700.160} 50 of 64
VERIFY DRC ..... Sub-Area : 50 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {485.760 1457.280 728.640 1700.160} 51 of 64
VERIFY DRC ..... Sub-Area : 51 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {728.640 1457.280 971.520 1700.160} 52 of 64
VERIFY DRC ..... Sub-Area : 52 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {971.520 1457.280 1214.400 1700.160} 53 of 64
VERIFY DRC ..... Sub-Area : 53 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1214.400 1457.280 1457.280 1700.160} 54 of 64
VERIFY DRC ..... Sub-Area : 54 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1457.280 1457.280 1700.160 1700.160} 55 of 64
VERIFY DRC ..... Sub-Area : 55 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1700.160 1457.280 1939.840 1700.160} 56 of 64
VERIFY DRC ..... Sub-Area : 56 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 1700.160 242.880 1939.840} 57 of 64
VERIFY DRC ..... Sub-Area : 57 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {242.880 1700.160 485.760 1939.840} 58 of 64
VERIFY DRC ..... Sub-Area : 58 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {485.760 1700.160 728.640 1939.840} 59 of 64
VERIFY DRC ..... Sub-Area : 59 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {728.640 1700.160 971.520 1939.840} 60 of 64
VERIFY DRC ..... Sub-Area : 60 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {971.520 1700.160 1214.400 1939.840} 61 of 64
VERIFY DRC ..... Sub-Area : 61 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1214.400 1700.160 1457.280 1939.840} 62 of 64
VERIFY DRC ..... Sub-Area : 62 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1457.280 1700.160 1700.160 1939.840} 63 of 64
VERIFY DRC ..... Sub-Area : 63 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1700.160 1700.160 1939.840 1939.840} 64 of 64
VERIFY DRC ..... Sub-Area : 64 complete 0 Viols.

Verification Complete : 30 Viols.

Violation Summary By Layer and Type:

      Short  MetSpC  Totals
M1         17       7      24
M2          5       1       6
Totals     22       8      30

*** End Verify DRC (CPU: 0:00:00.5 ELAPSED TIME: 0.00 MEM: 14.0M) ***

innovus 1> █
```

```
File Edit View Search Terminal Help
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d01.5' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d01.6' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d01.7' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d01.8' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d01.9' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d01.10' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d01.11' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d01.12' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d01.13' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d01.14' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d05.1' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d05.2' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d05.3' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (IMPVL-520): P/G pin 'VSS0' of instance 'pc3d05.4' is not connected to a power or ground net. Use globalNetConnect to connect it to a power or ground net. In a CPF-based flow, you can also use a create global connection command in the CPF to specify the connection.
Type 'man IMPVL-520' for more detail.
**WARN: (ENS-27): Message (IMPVL-520) has exceeded the current message display limit of 20.
To increase the message display limit, refer to the product command reference manual.
Pwr name (VDD CORE).
Pwr name (VDD CORE).
Gnd name (VSS CORE).
2 Pwr names and 1 Gnd names.
innovus 3> saveNetList ./Signoff/fifo_signoff_withP.v
Writing NetList ./Signoff/fifo_signoff_withP.v ...
innovus 4> write_sdc ./Signoff/fifo_signoff.sdc
innovus 5> █
```



# Signoff and save files

```
File Edit View Search Terminal Help
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=1500.6M) ***
#Saving pin access data to file Signoff/fifo.signoff.enc.dat.tmp/fifo.apa ...
% Begin Save power constraints data ... (date=10/30 19:55:37, mem=1200.2M)
% End Save power constraints data ... (date=10/30 19:55:37, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.2M, current mem=1200.2M)
Generated self-contained design fifo.signoff.enc.dat.tmp
% End save design ... (date=10/30 19:55:38, total cpu=0:00:00.6, real=0:00:12.0, peak res=1230.9M, current mem=1200.6M)
*** Message Summary: 0 warning(s), 0 error(s)

The in-memory database contained RC information but was not saved. To save
the RC information, use saveDesign's -rc option. Note: Saving RC information can be quite large,
so it should only be saved when it is really desired.
% Begin save design ... (date=10/30 19:55:38, mem=1200.6M)
% Begin Save ccopt configuration ... (date=10/30 19:55:38, mem=1200.6M)
% End Save ccopt configuration ... (date=10/30 19:55:38, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
% Begin Save netlist data ... (date=10/30 19:55:38, mem=1200.6M)
Writing Binary DB to Signoff/fifo.signoff.enc.dat.tmp/fifo.v.bin in single-threaded mode...
% End Save netlist data ... (date=10/30 19:55:38, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
Saving symbol-table file ...
Saving congestion map file Signoff/fifo.signoff.enc.dat.tmp/fifo.route.congmap.gz ...
% Begin Save AAE data ... (date=10/30 19:55:38, mem=1200.6M)
Saving AAE data ...
% End Save AAE data ... (date=10/30 19:55:38, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
Saving preference file Signoff/fifo.signoff.enc.dat.tmp/gui.pref.tcl ...
Saving mode setting ...
Saving global file ...
% Begin Save floorplan data ... (date=10/30 19:55:43, mem=1200.6M)
Saving floorplan file ...
% End Save floorplan data ... (date=10/30 19:55:43, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
Saving PG file Signoff/fifo.signoff.enc.dat.tmp/fifo.pg.gz, version#2, (Created by Innovus v20.14-s095.1 on Thu Oct 30 19:55:43 2025)
*** Completed savePofFile (cpu=0:00:00.0 real=0:00:01.0 mem=1521.3M) ***
Saving Drc markers ...
... 166 markers are saved ...
... 58 geometry drc markers are saved ...
... 20 antenna drc markers are saved ...
% Begin Save placement data ... (date=10/30 19:55:44, mem=1200.6M)
** Saving stdCellPlacement binary (version# 2) ...
Save Adaptive View Pruning View Names to Binary file
% End Save placement data ... (date=10/30 19:55:44, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
% Begin Save routing data ... (date=10/30 19:55:44, mem=1200.6M)
Saving route file ...
*** Completed saveRoute (cpu=0:00:00.0 real=0:00:00.0 mem=1521.3M) ***
% End Save routing data ... (date=10/30 19:55:44, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
Saving property file Signoff/fifo.signoff.enc.dat.tmp/fifo.prop
*** Completed saveProperty (cpu=0:00:00.0 real=0:00:00.0 mem=1524.3M) ***
#Saving pin access data to file Signoff/fifo.signoff.enc.dat.tmp/fifo.apa ...
% Begin Save power constraints data ... (date=10/30 19:55:44, mem=1200.6M)
% End Save power constraints data ... (date=10/30 19:55:44, total cpu=0:00:00.0, real=0:00:00.0, peak res=1200.6M, current mem=1200.6M)
Generated self-contained design fifo.signoff.enc.dat.tmp
% End save design ... (date=10/30 19:55:44, total cpu=0:00:00.6, real=0:00:06.0, peak res=1231.4M, current mem=1200.7M)
*** Message Summary: 0 warning(s), 0 error(s)
```

```
File Edit View Search Terminal Help
Number of Input/InOut Floating Pins      : 2
Number of Output Floating Pins           : 0
Number of Output Term Marked TieHi/Lo    *: 0

**WARN: (IMPREPO-216): There are 20 Instances with input pins tied together.
**WARN: (IMPREPO-218): There are 2 Floating Instance terminals.
Number of nets with tri-state drivers     : 0
Number of nets with parallel drivers       : 0
Number of nets with multiple drivers       : 0
Number of nets with no driver (No FanIn)   : 0
Number of Output Floating nets (No FanOut) : 3
Number of High Fanout nets (>50)           : 3
**WARN: (IMPREPO-227): There are 3 High Fanout nets (>50).
Checking routing tracks.....
Checking other grids.....
Checking FINFET Grid is on Manufacture Grid.....

Checking core/die box is on Grid.....

WARNING (IMPFP-7238): CORE's corner: (374.88000000000000, 374.88000000000000) is NOT on PlacementGrid, Please use command getFPlanMode to check current Grid settings, use command setFPlanMode to change which grid to s
nap to. And use command get_snap_grid.info to get grids' offset and pitch. Command floorplan can be used to fix this issue.
WARNING (IMPFP-7238): CORE's corner: (1564.96000000000000, 1564.96000000000000) is NOT on PlacementGrid, Please use command getFPlanMode to check current Grid settings, use command setFPlanMode to change which grid to
snap to. And use command get_snap_grid.info to get grids' offset and pitch. Command floorplan can be used to fix this issue.
Checking snap rule .....

Checking Row is on grid.....

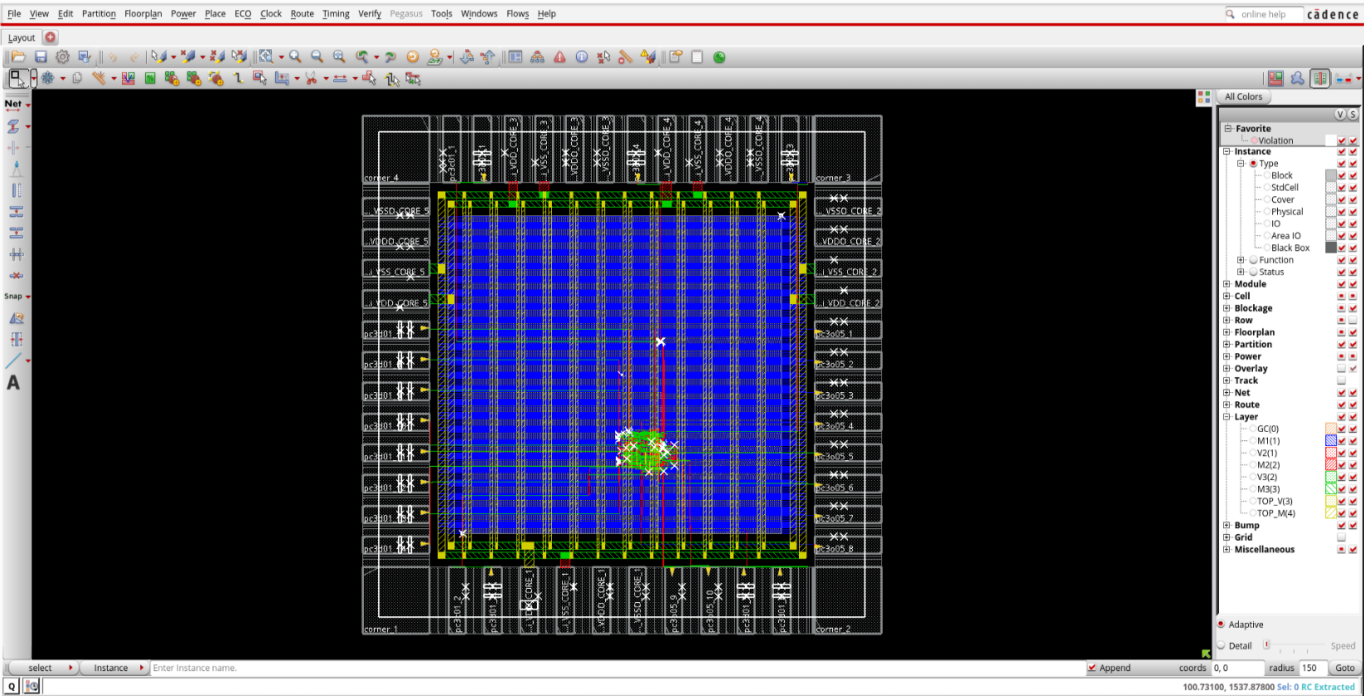
Checking AreaIO row.....
Checking routing blockage.....
Checking components.....
Checking constraints (guide/region/fence).....
Checking groups.....
Checking Pin Core Box.....

Checking Preroutes.....
No. of regular pre-routes not on tracks : 0
Design check done.
Report saved in file checkDesign/fifo.main.htm.ascii

*** Summary of all messages that are not suppressed in this session:
Severity ID Count Summary
WARNING IMPDB-2136 2 2 %s term '%s' of %s instance '%s' does no...
WARNING IMPREPO-227 1 There are %d High Fanout nets (>50).
WARNING IMPREPO-205 1 There are %d cells with missing PG PIN.
WARNING IMPREPO-216 1 There are %d Instances with input pins t...
WARNING IMPREPO-218 1 There are %d Floating Instance terminals...
*** Message Summary: 6 warning(s), 0 error(s)

Innovus S> The in-memory database contained RC information but was not saved. To save
the RC information, use saveDesign's -rc option. Note: Saving RC information can be quite large,
so it should only be saved when it is really desired.
% Begin save design ... (date=10/30 19:55:14, mem=1191.7M)
% Begin Save ccopt configuration ... (date=10/30 19:55:14, mem=1193.7M)
```

# GDS Ready



| Object            | Count |
|-------------------|-------|
| Instances         | 50788 |
| Ports/Pins        | 0     |
| Nets              | 5209  |
| metal layer M1    | 766   |
| metal layer M2    | 2994  |
| metal layer M3    | 1416  |
| metal layer TOP_M | 24    |
| Via Instances     | 2725  |
| Special Nets      | 701   |
| metal layer M1    | 653   |
| metal layer M2    | 11    |
| metal layer M3    | 0     |
| metal layer TOP_M | 29    |
| Via Instances     | 1336  |
| Metal Fills       | 0     |
| Via Instances     | 0     |
| Metal FillOPCs    | 0     |
| Via Instances     | 0     |
| Metal FillDRCS    | 0     |
| Via Instances     | 0     |
| Text              | 434   |
| metal layer M1    | 123   |
| metal layer M2    | 257   |
| metal layer M3    | 54    |
| Blockages         | 0     |
| Custom Text       | 0     |
| Custom Box        | 0     |
| Trim Metal        | 0     |

#####Streamout is finished!  
innovus 3>

